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Boron-doped Carbon Microspheres as the Catalyst for Rechargeable Al-air Batteries

Tirong Guo^a, Xue Qin^{a,b,*}, Xuejiao Gao^a, Lina Hou^a, Jia Li^a, Xiang Li^a,
Tao Lei^c, Jun Lv^c

^aDepartment of Chemistry, School of Science, Tianjin University, and Collaborative Innovation Center of Chemical Science and Engineering (Tianjin), Tianjin 300072, China

^bKey Laboratory of Advanced Energy Materials Chemistry (Ministry of Education), College of Chemistry, Nankai University, Tianjin 300071, China

^cJiangsu GE New Energy Technology Co., Ltd., Jiangsu 212200, China

*Corresponding Author E-mail Address : qinxue@tju.edu.cn

Telephone Number : +86-13920603736

Abstract

Three-dimensional mesoporous boron-doped carbon microspheres (B-CMs) were synthesized through a facile impregnation-calcination method. The B atoms introduced plentiful defects into the surfaces of CMs and were mainly presented in the form of boron carbide, which increases the specific surface area of the composite material and endows it with enhanced catalytic activity. A rechargeable aluminum (Al)-air battery with B-CMs catalysts exhibited low overpotentials and its effective discharge capacity reached 300 mAh g⁻¹ at the current density of 50 mA g⁻¹. The capacity of Al-air battery was maintained at 584 mAh g⁻¹ after 30 cycles.

Keywords: Al-air battery; Boron-doped carbon; Ionic liquid; Rechargeable; Cyclic performance;

1. Introduction

With the urgent demanding of energy, lithium-ion batteries (LIBs) with low energy density are unable to meet the increasing energy requirements of various electrical devices. Tremendous investments of effort have thus been made in expanding the range of energy markets. Metal-air batteries, as one rather promising research domain, have aroused extensive attention for decades based on their great potential to deliver high energy density and capacity accessibility. Lithium-air batteries (LABs), though, has compensated for this shortcoming because of the intrinsic largest energy density in the metal-air cells' family, the availability and high

cost of Li metal still present drawbacks. As for other types of metal-air batteries, Al-air batteries, ranking the second in terms of energy density, just lower to that of Li, are among the most developed battery systems. Besides, Al is the topmost metal element in content in the earth's crust with a significantly lower cost. It has a high-volume charge capacity (8.0 Ah cm^{-3} for Al, 2.06 Ah cm^{-3} for Li) and its weight charge capacity is comparable to that of Li (3.0 Ah g^{-1} for Al, 3.9 Ah g^{-1} for Li) [1-5].

Whereas, the self-corrosion and surface passivation of Al is the major obstacle in traditional aqueous Al-air batteries. In order to hamper the self-discharge reaction of Al effectively, one possible route is to implement non-aqueous solvents as substitutes. Aprotic electrolytes applied in Li-air batteries, such as organic carbonates, ethers and esters, are not efficient in dissolving the surface oxide layer and activating the Al anode, thus are unsuitable in the Al-air cells [6,7]. Despite there are several innovative findings about these batteries with ionic liquid electrolytes and polymer solid-state electrolytes, which shed some light in this field, more detailed and in-depth investigations are of necessity in the following years [8]. Previously, multiple works have reported the application of room temperature ionic liquids (RTILs) in rechargeable aluminum batteries [9,10]. The acidic $\text{AlCl}_3/[\text{EMIm}]\text{Cl}$ has been demonstrated as an efficient electrolyte, providing a promising dawn for rechargeable Al-air batteries [11]. Thus, $\text{AlCl}_3/[\text{EMIm}]\text{Cl}$ was also used as a reliable electrolyte for the Al-air battery in this work.

Among the wide variety of catalysts, carbon-based materials have attracted much attention due to their high surface area, good stability and excellent electrical activity [12-15]. Numerous studies have indicated that boron-doped carbon materials not only enlarge the specific surface area, but also increase the number of active sites for oxygen absorption, thus improving the catalytic efficiency of oxygen reduction reaction (ORR) [16,17]. However, research on cathode catalysts for non-aqueous Al-air batteries is still in its infancy. In this study, B-CMs were synthesized by a simple impregnation-calcination method to improve the overall electrochemical performance of non-aqueous Al-air cells.

2. Material and methods

2.1. Catalytic material preparation

B-CMs were synthesized as follows: sucrose (6.3 g) and 6.0 M diluted sulfuric acid (120.0 g) were mixed and heated to reflux for 10 hours. The obtained product was washed, dried and ground finely. This precursor (1.0 g) and a saturated boric acid solution (20.0 g) were mixed and stirred for another 10 h. The resulting suspension

was washed with distilled water twice, then dried at 60 °C and pestled into fine powder. We obtained B-CMs by calcining the powder in a nitrogen atmosphere at 1,000 °C for 2 hours and activating it in ambient air at 400 °C for 1 hour.

2.2. Material characterization

The surface morphology and microstructures were studied using transmission electron microscopy (TEM) (G2 F20; Tecnai) and field emission scanning electron microscopy (FE-SEM) (S-4800; Hitachi). The crystal structures of the samples were characterized by X-ray diffraction (XRD) (D/MAX-2500; Rigaku) and Raman spectroscopy (InVia Rflex; Renishaw). The Brunauer-Emmet-Teller (BET) method was utilized to calculate the specific surface areas of the samples.

2.3. Battery assembly and electrochemical measurement

The electrochemical behaviors of the samples were studied by using Swagelok-type cells. The slurry was uniformly deposited on a piece of carbon paper by mixing B-CMs, super-p and polyvinylidene fluoride (PVDF) with a mass ratio of 7:2:1 in the N-methyl-2-pyrrolidone (NMP) solvent (a mass loading of 0.50 mg cm⁻²). An Al foil was used as the anode and a polytetrafluoroethylene (PTFE) membrane was applied as the separator. The electrolyte was derived from mixing AlCl₃ and 1-ethyl-3-methylimidazolium chloride ([EMIm]Cl) with a molar ratio of 1.3:1. After the mounting process, batteries were kept to rest for 5 hours before the activation of tests. The cells were tested by a battery testing system (LAND-CT2001A system). Cyclic voltammetry (CV) curves were collected using a Zahner Ennium electrochemical workstation.

3. Results and Discussion

Fig. 1(a) is the Raman test of the synthesized composites. It can be seen that the D-peak (1,359 cm⁻¹) and the G-peak (1,600 cm⁻¹) of CMs are not affected by the addition of the B atoms. Meanwhile, the value of I_D/I_G has increased for B-CMs, thus indicating that the introduced B atoms created more defects on the surfaces of CMs. Notably, B-CMs exhibit additional peaks at 2,707 cm⁻¹ and 2,909 cm⁻¹, which further illustrates that B-CMs have been prepared successfully [18,19]. From the XRD results (Fig. 1(b)), the two strong peaks (25.6° and 42.5°) are attributed to characteristic peaks of carbon; while these two peaks become sharper and more prominent for B-CMs compared with bare CMs, meaning a higher degree of crystallinity structure in B-CMs. The diffraction peaks of boron carbide (B_xC_y) are obvious in the spectra of B-CMs, signifying that the B atoms were supported on CMs in the form of B_xC_y, which can activate the π electrons of carbon and act as active sites for ORR and OER

[20-22].

Fig. 1(c)-(f) are the surface morphology and microstructure of pure CMs and B-CMs composites. It can be observed that the surfaces of B-CMs are covered with white structure layers at lower magnification, which are proved to be uneven surfaces of the samples when magnified at high resolution of TEM. No other obvious distinction exists between these two samples in the diameters of the microspheres, both of which are about ~ 400 nm. One can also find that the B doping has caused the surfaces of CMs to become rough with more defects due to etching by the B atoms, while the overall spherical shape integrity is well-maintained from the above SEM and TEM images. The specific surface area and pore diameters of the synthesized materials were characterized and calculated via BET method. As shown in Fig. 1(g)-(h), the two samples exhibit a typical type-IV N_2 adsorption-desorption isotherm with a H3 hysteresis loop. The specific surface area of the B-CMs reaches $832.9 \text{ m}^2 \text{ g}^{-1}$, which is higher than that of the CMs with $692.5 \text{ m}^2 \text{ g}^{-1}$. The pore size distributions of both samples are centered at 2.0 nm - 5.0 nm. The larger specific surface area and mesoporous structure of B-CMs are not only favorable of facilitating the electrolyte infiltration and ion transport, but can also promote the oxygen diffusion process.

The electrochemistry properties of non-aqueous Al-air batteries with CMs and B-CMs were investigated. The batteries were tested in the voltage range of 0.2 V – 2.5 V. Figure 2(a) shows the distinction between the first charge-discharge voltages of CMs and B-CMs with a cutoff capacity of 250 mAh g^{-1} at a current density of 50 mA g^{-1} . B-CMs exhibited lower charge voltage and higher discharge voltage, while CMs showed a wide voltage gap. This may be resulted from the nature of slow chemical kinetics for pure carbon materials. Once heteroatoms B were introduced and associated with C atoms, the electrical neutral was broken due to the conjugation between the B empty spin-orbitals and the original inert electrons of C. The positively charged B atoms could act as catalytic active sites, rendering the B-CMs with higher catalytic activity and therefore significantly diminishing the overpotential. As shown in Fig. 2(b), two reduction peaks of CMs were located at 0.7 V and 1.3V, while a clear reduction peak at about 1.5 V and a weak reduction peak at about 0.73 V of B-CMs can be seen. Besides, there were no obvious oxidation peaks for CMs in the OER process, which contrasts sharply with the conspicuous oxidation peak located at about 2.0 V for B-CMs. Furthermore, B-CMs exhibited markedly higher peak current and larger peak area during both ORR and OER. These results again elucidate that B-CMs are effective bifunctional catalysts with greatly enhanced electrocatalytic activity and

good reversibility in non-aqueous Al-air cells.

When the charge capacity was limited to 1000 mAh g⁻¹ at a current density of 50 mA g⁻¹ (Fig. 2(c)), the discharge capacity of the battery with B-CMs could deliver 773.5 mAh g⁻¹ in the first cycle followed by stable operations over 30 cycles, with the discharge platform between 1.5 V - 1.8 V. The effective discharge capacity in this voltage range reached almost 300 mAh g⁻¹. The evident numerical advantage in the first cycle discharge capacity of the B-CMs, comparing to the following cycles, might be related to the scanty rest time prior to the initial of test. It has been reported that the immersion of Al in ionic liquid electrolyte could dissolve the surface oxide layer and form a new stable layer by the corrosion reaction of Al anode with the electrolyte [7]. Meaning, the discharge with an insufficient rest period might be accompanied by the regular ORR process and the spontaneous self-discharge phenomenon. The function of this protective layer might be similar to the SEI layer in the Li-ion battery systems by the reductive decomposition of electrolyte [23-25]. As is protected by the stable layer, the subsequent cycles are able to be maintained with almost overlapping curves, as shown in Fig. 2(c). Moreover, it is obvious that the discharge process consisted of two stages (Fig. 2(c)), namely the steady-discharge plateau in the first stage and the gently descending curve in the second stage. Gao et al. [26] have proposed the following reaction mechanism for the discharge process:



which is highly consistent with the two reduction peaks in the CV curves as presented before (Fig. 2(b)). The two discharge stages appearing on the discharge curves are also corresponded to the reaction routes shown in reversible reaction equations (1) and (2). The long term cycle life of Al-air batteries with different materials was depicted in Fig. 2(d). It can be seen clearly that both of the batteries followed the same trend, reaching the summit in the first cycle of discharge, then dipping to the relatively stable discharge stages. This might be ascribed to the short rest period and the interaction of Al with the ionic liquid electrolyte discussed above. Additionally, the discharge capacity for the battery with B-CMs remained stable over 30 cycles with a discharge capacity of 584 mAh g⁻¹ at the current density of 50 mA g⁻¹, while that for the battery with CMs decayed sharply to a rather low level. This demonstrates that B-CMs can be considered as a promising candidate for cathode catalysts with an excellent catalytic service lifespan.

To evaluate the chemical nature of the discharge products, elemental mapping and EDX tests were performed with the discharged cathode (Fig. 3(a)-(b)). From the close distribution of the material elements in the blue and yellow marked boxes, it can be inferred that these large particles consist mainly of Al, Cl, and O. Very early on, D. Gelman and his coworkers had determined that the discharge product was Al_2O_3 [3,27]. The suggestion by Gao et al. [26] that Al_2O_3 was one of the products generated during the discharge reaction is in good agreement with our elemental detection results. Hence, it is reasonable to assume that the main discharge product is Al_2O_3 . Our preliminary work might provide perspectives and basis for the future intensive researches of this field.

4. Conclusion

In this study, B-CMs composites with high catalytic performance were synthesized by an easily reliable method. The results of XRD and Raman spectroscopy show that the B atoms were mainly in the form of boron carbide and introduced many defects on the surfaces of CMs. The increased specific surface area and the doped B atoms promised better performance both in ORR and OER of the non-aqueous Al-air batteries with the composite catalysts. The Al-air battery with B-CMs exhibited an effective discharge capacity of 300 mAh g^{-1} at the discharge stage between 1.5 V - 1.8 V with the current density of 50 mA g^{-1} . The discharge capacity still remained at 584 mAh g^{-1} after 30 cycles.

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Declarations of interest: none.

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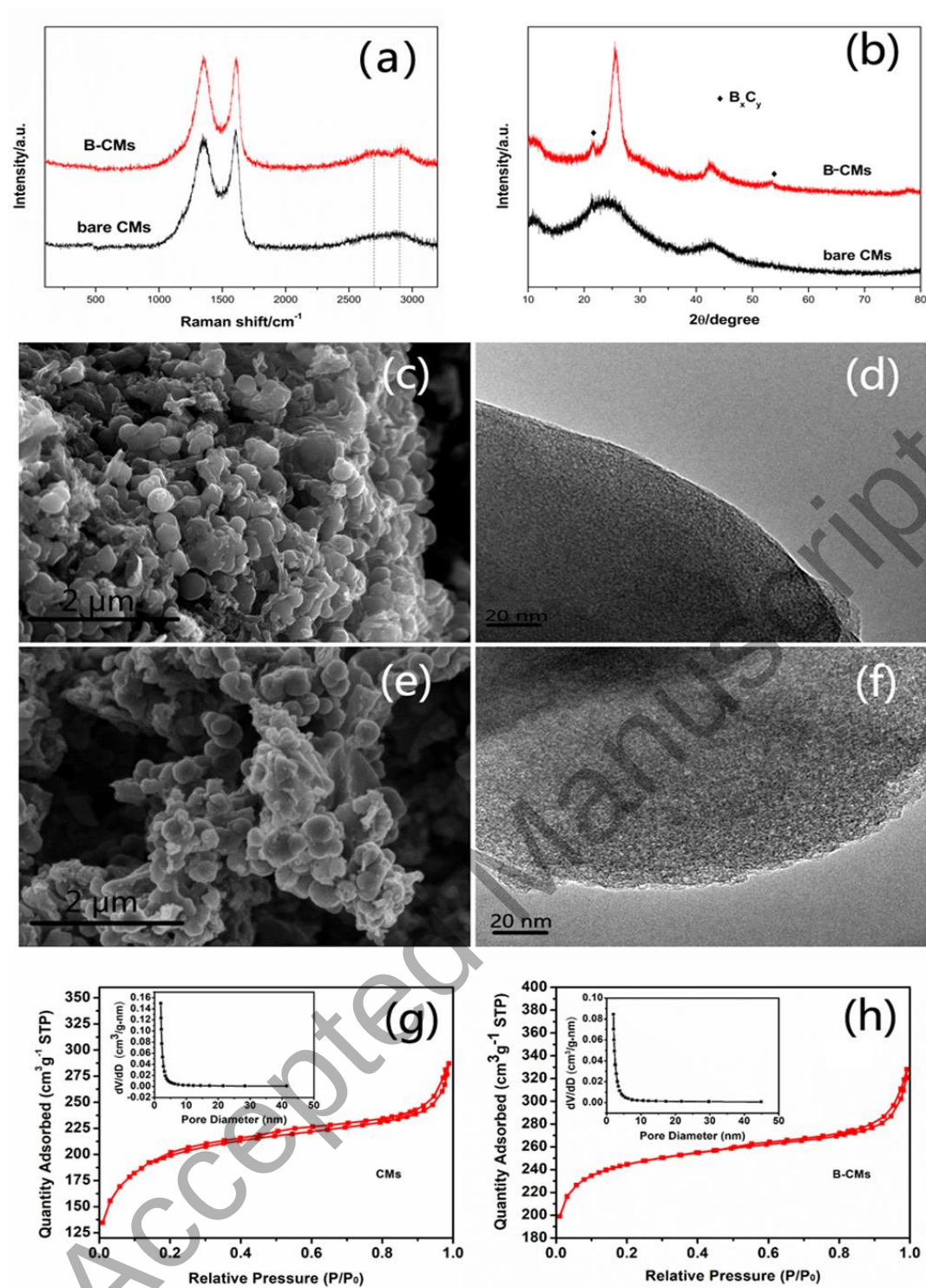


Fig. 1. (a) Raman diagrams (b) XRD patterns of bare CMs and B-CMs; SEM and TEM images of (c) (d) bare CMs and (e) (f) B-CMs; (g) (h) N₂ adsorption-desorption isotherms of CMs and B-CMs.

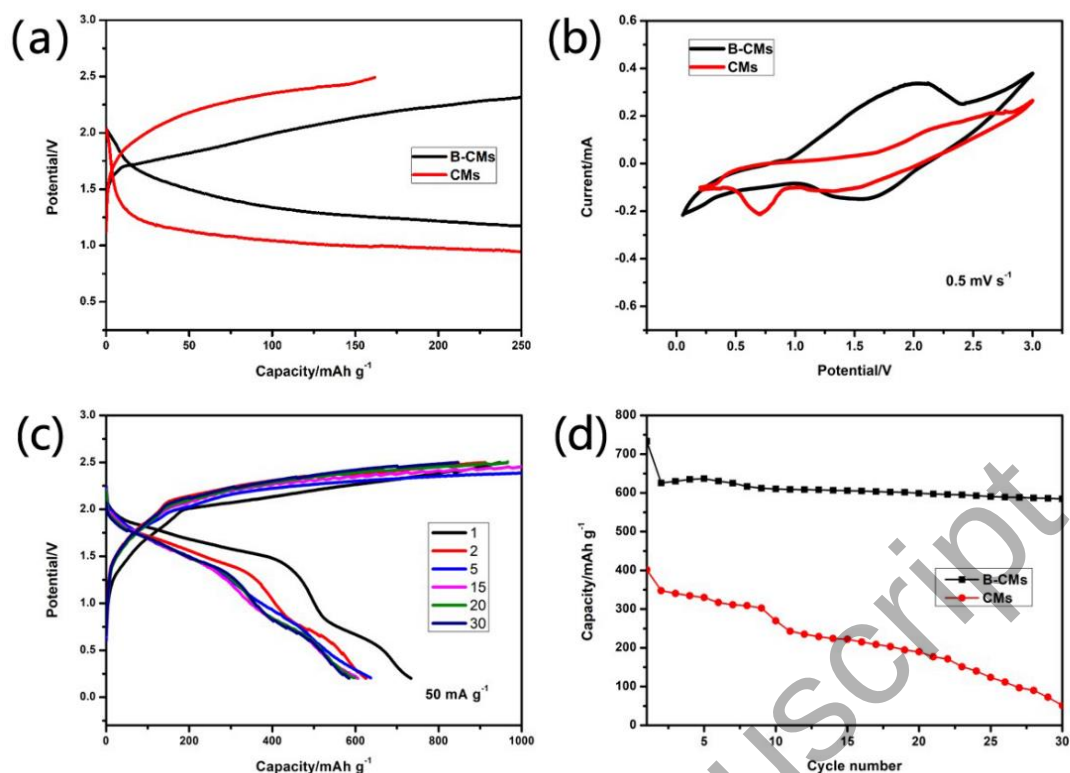


Fig. 2. (a) The first charge-discharge profiles of CMs and B-CMs with a cutoff capacity of 250 mAh g⁻¹ at a current density of 50 mA g⁻¹; (b) The CV curves at a scan rate of 0.5 mV s⁻¹; (c) The 10th cycle charge-discharge performance of B-CMs with a charge capacity of 1000 mAh g⁻¹, and (d) the comparison of discharge life curves both at the current density of 50 mA g⁻¹.

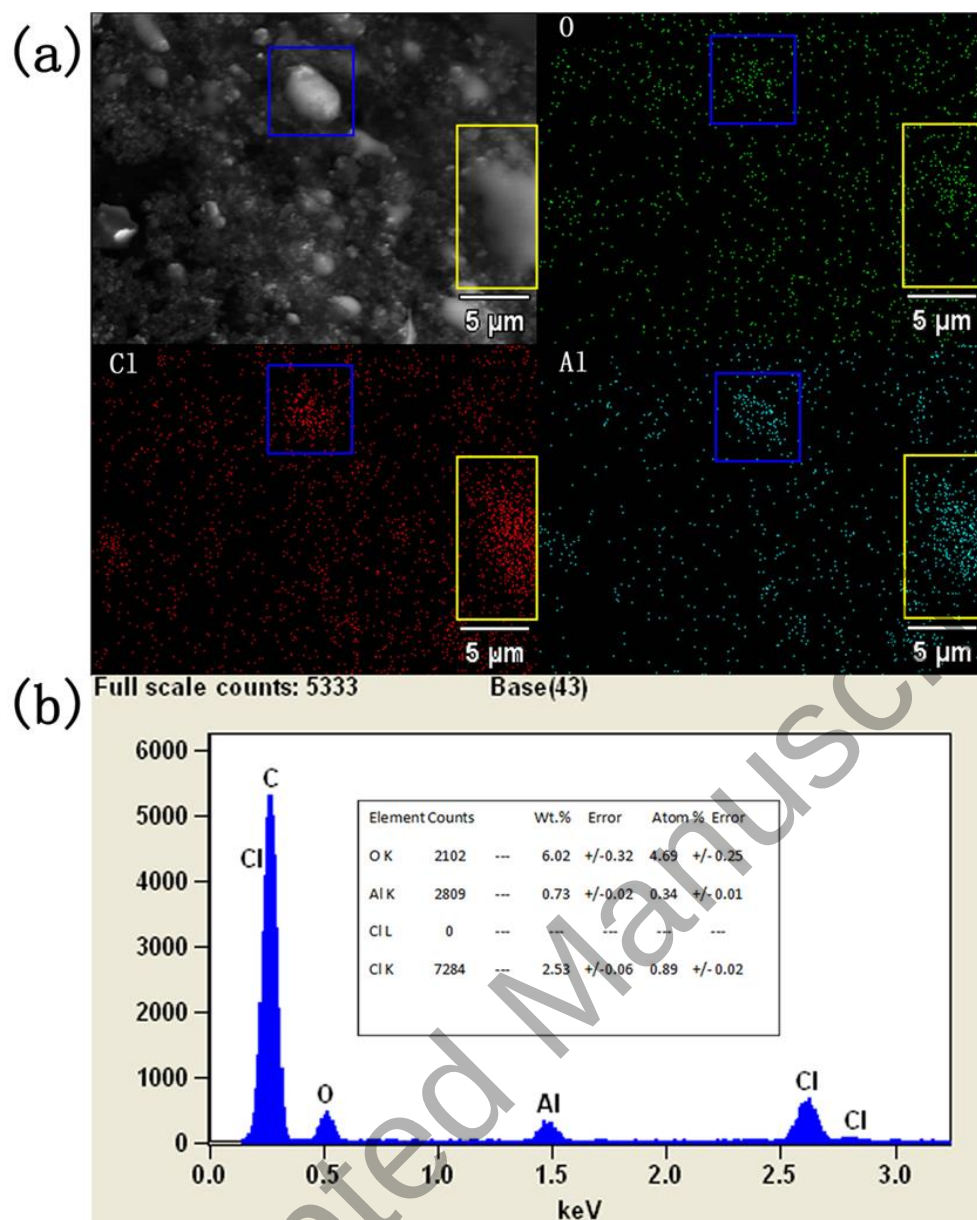


Fig. 3. (a) Elemental mapping images and (b) EDX analysis of the air cathode with B-CMs as catalyst after discharge.